Samsung Gen AI Hackathon 2025

Climb the corporate ladder

Welcome to the future of public speaking training.

Team

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Submission:

Google drive link with Unity exe file and demo video –

https://drive.google.com/drive/folders/1c-oq_U8_P13tZrMTBVBMZl3De_2TbDOE?usp=sharing

Google drive link with Unity game files –

https://drive.google.com/file/d/17cXMqM8OWzklj_dTKQJ-P-MVApiHzuqm/view?usp=sharing

What is Climb the corporate ladder?

Climb the corporate ladder is a Gen AI – Powered gamified public speaking training experience with multiple levels, challenges and minigames.

An experience which is going to transform even the most scaredy of cats into a confident, well-spoken and educated professional ready to face the challenges in today's modern world.

Who is it for?

It's for those with great talent and intelligence, but who lack the confidence to speak up in high pressure scenarios such as job interviews, networking events, and public speaking. This ensures that these individuals do not face setbacks on their career growth or personal success.

We target this young and ever-growing workforce entering the job market with less than ideal communication skills but a drive to improve. From urban cities to rural regions, every individual will have the chance to compete in the market on equal terms.

What makes us different?

The existing confidence building programs are rigid, one-size-fits-all workshops or online courses that fail to adapt to individual needs. They tend to be repetitive, theoretical, and disengaging, with little room for personalization or real-time feedback. As a result, users often lose interest quickly and struggle to translate what they learn into genuine, lasting behavioral change.

There is a clear need for a more engaging, adaptive, and gamified approach to confidence building. By combining interactive storytelling, AI driven feedback, and real world practice scenarios, we can create a safe yet immersive environment where users learn by doing. This opens the door for a transformative solution that not only builds confidence but sustains it through continuous growth and personalized guidance.

Highlights:

- Leverages LangGraph AI Agents for dynamic, context-aware verbal/non-verbal feedback
- Proprietary AI confidence scoring ensures measurable, engaging progress
- Evaluates tone, clarity, and assertiveness, helping users practice communication in realistic scenarios.
- Users are analyzed for body language, facial expressions, and posture using immersive video integration. The system provides actionable insights on non-verbal presence and communication style, helping users refine both confidence and delivery in real-life scenarios.

How does it work?

User Experience :

The game is developed using the Unity game engine.

Explores Immersive environment based on Real life scenes set in Outside world, Office environment, large presentation hall.

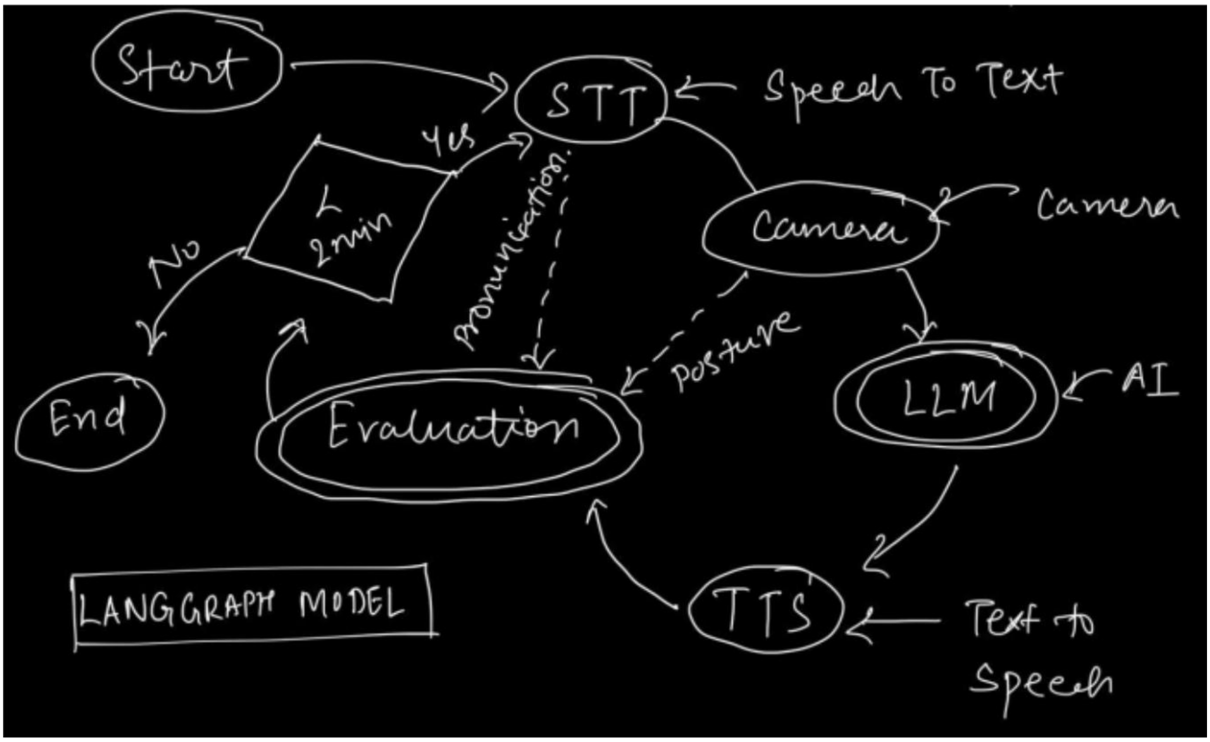
Interact with Dynamic AI Characters in each environment. NPCs with unique personalities are customized with individual memory based on your interactions adding a depth of realism to the game

Interactive minigames to improve upon English skills like Hangman and many more to come.

Our game uses a Langgraph based architecture with Google Gemini as the llm of choice.

One of our great strengths are our conversation based minigames with a fixed goal in mind (convince the boss, try to engage an audience during a presentation).

For these games, the following is the structure of the graph-



As you can see,

1)the graph starts by receiving some audio from the frontend.

2)Then using OpenAi whisper, the audio gets transcribed.

3)Then using camera and mediapipe, your posture is examined and this data is added to your text.

4) After this, the data gets sent to the llm, which replies in character.

5)Then using streaming and coqui-TTS the audio of the reply gets played back to the user.

6)Then the reply along with your audio is sent to an evaluation node. This node examines your conversation and passes or fails you impacting your pass meter.

7) This data is added to your state and passed in the next turn. Based on this data, the AI will either get more or less aggressive with you.

8)If a certain time limit has passed, your final evaluation triggers. Else your conversation continues.

9)In your final evaluation, If you have a positive pass meter, you passed! else you need to try again.

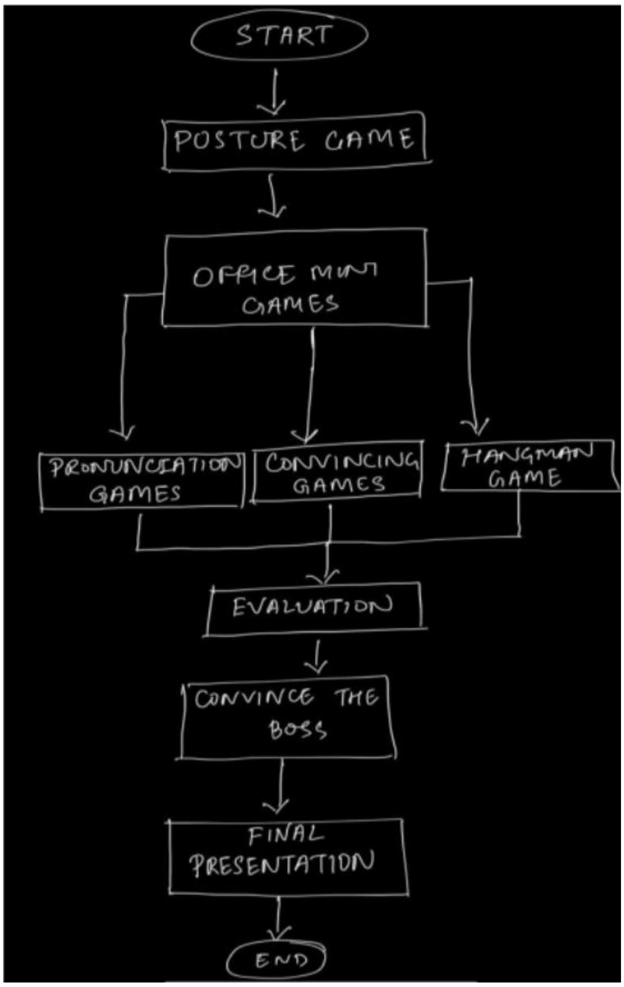
For our current demo, we have 3 LangGraph endpoints:

1. Office NPCs – make the office environment more interactive and lively.
2. Office Boss – drives the "Convince the Boss" mini-game.
3. Audience – reacts and engages during the Final Presentation.

Posture Estimation Node

- Use OpenCV and MediaPipe to extract body mesh points
- Calculate slouching angle or looking away
- Determine if the user is slouched or attentive

Flow of the game:



Feasibility:

- Technically Feasible: Leverages existing open-source AI tools and Unity's mature ecosystem for quick development
- Cost-effective: Free tiers of AI APIs for prototyping, Open-source alternatives reduce expenses
- Market Ready: Aligns with rising demand for EdTech gamification

Potential Challenges and Risks

- Privacy concerns with voice/video data: Handling sensitive user audio and video recordings requires robust data protection measures and clear privacy policies
- AI accuracy issues in diverse accents and languages: Current models may struggle with regional accents, dialects, and non-native English speakers, affecting feedback quality

- High development time for branching storylines: Creating multiple narrative paths and character interactions significantly increases content creation workload
- Dependency on external APIs: Reliance on third-party services introduces potential downtime risks and service availability issues

Current Limitations

One of our current technical limitations is the use of local Speech-to-Text (STT) and Text-to-Speech (TTS) models, which are computationally heavy for a real-time gaming environment. These local models impact performance and require significant system resources.

We plan to upgrade to hosted API models for STT and TTS services if we secure additional resources, which would improve performance, reduce local computational requirements, and provide more reliable service delivery.

What do you need to run it-

1. Exe game file
2. Google Gemini Api.
3. OpenAI Whisper
4. Coqui-TTS
5. Github Code

How to run our game

1.Download the code from github as a .zip file.

2. UnZip it and place it in a folder.

3. In a terminal navigate to the folder you placed the game in and create a virtual environment by typing **python -m venv .venv** .

4. Activate the virtual environment by typing **.venv/Scripts/activate**

5. Next type **pip install -r requirements.txt**

6.Go to https://aistudio.google.com/ and click on get API key. Then follow the instructions to create a free API key.

7.Create a .env file in the game folder and paste your api key there in the format GOOGLE_API_KEY="YOUR API KEY HERE "

8. Download OpenAI Whisper from https://github.com/openai/whisper .

9.Type **uvicorn backend:app --host 0.0.0.0 --port 8000 -**reload in terminal to start the backend server.

10. In another terminal, navigate to the folder where you placed the game in and type **python nre.py**

11. Click on the .exe file to start your game and enjoy.